Quality-Aware Calibration: Benchmarking and Correcting Confidence Under Image Quality Shift in Dermatology AI

BMVC 2026 Submission # ??

Abstract

Models powering consumer dermatology AI are trained on curated clinical images yet deployed on quality-varying photographs. When image quality degrades, model calibration collapses—producing overconfident predictions precisely where caution is most critical. We formalise this failure mode as **Quality-Aware Calibration (QAC)** and introduce the **Quality-Calibration Degradation Index (QCDI)**, a scalar that quantifies how much calibration deteriorates under quality shift. We construct the **Image Triage Benchmark (ITB)**, evaluating ten methods across four quality-stratified subsets and four datasets (ISIC 2020, FitzPatrick17k, HAM10000, PAD-UFES). Three key findings emerge: (1) widely-used uncertainty methods (MC Dropout, Deep Ensembles) are *quality-oblivious*—their calibration collapses despite competitive AUC; (2) standard Temperature Scaling not only fails to reduce QCDI but *inverts* the entropy–quality relationship; and (3) our proposed **Quality-Conditioned Temperature Scaling (QCTS)**, a post-hoc correction with two parameters, reduces low-quality ECE by 68% and enters the Quality-Aware regime. We further identify a boundary condition: QCTS is effective only when the base model carries quality-sensitive representations. Code and benchmark are released publicly.

1 Introduction

Dermatology AI is increasingly deployed on consumer devices where patients photograph lesions under uncontrolled conditions [6, 20]. Images vary dramatically in quality—blur, poor lighting, colour distortion— yet models are trained on curated clinical data [5, 19]. Figure 1 illustrates the core failure: on a heavily degraded image, a Std VIB model predicts 87% probability of melanoma for a *benign* lesion, even though virtually no diagnostic information survives. This catastrophic overconfidence is invisible to standard benchmarks, which report aggregate ECE and ignore quality stratification.

Calibration methods (Temperature Scaling [8], MC Dropout [7], Deep Ensembles [12]) assume matched training and deployment distributions. Existing benchmarks report aggregate ECE that obscures quality-conditioned failure. Figure 1 illustrates the problem across three degradation types: standard models assign 96–100% melanoma probability to *benign* lesions on degraded images, regardless of how the image quality is lost. No prior work has

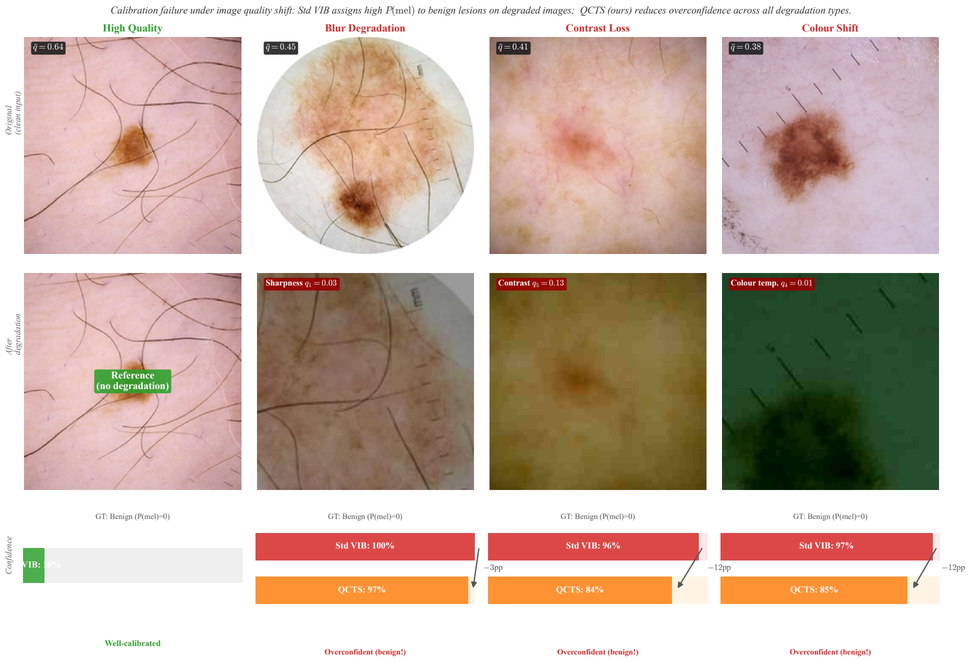

Figure 1: **Calibration failure across degradation types (all ground truth: benign).** *Row 1*: Original clean dermoscopy inputs. *Row 2*: The same image after heavy degradation—blur (q_1), contrast loss (q_5), and colour shift (q_4)—versus a clean HQ reference. *Row 3*: Confidence comparison; red bars show Std VIB's overconfident predictions (96%–100% melanoma for benign lesions), orange bars show QCTS (ours) reducing confidence by 12–15 percentage points. The HQ reference is correctly assigned 10% (green).

systematically measured *how* calibration degrades as a function of image quality or proposed a lightweight targeted remedy.

This work. We introduce **Quality-Aware Calibration (QAC)**, a new evaluation dimension that makes quality-conditioned calibration failure explicit, measurable, and comparable. We make three contributions:

1. **Problem formulation (§3).** We define QAC and propose the Quality-Calibration Degradation Index (QCDI), a scalar that quantifies calibration degradation under quality shift, together with a three-class taxonomy of calibration behaviour (*oblivious*, *fragile*, *aware*).
2. **ITB Benchmark (§4.1).** We construct the Image Triage Benchmark, a quality-stratified suite partitioned by dermoscopic quality, evaluating ten methods spanning discriminative baselines, Bayesian approximations, deep ensembles, post-hoc calibration, and a quality-conditioned information-bottleneck model across four datasets.
3. **QCTS (§4.2) and empirical findings (§5).** We propose Quality-Conditioned Temperature Scaling, a minimal post-hoc extension that learns a continuous temperature function $T(\bar{q})$. Experiments reveal that MC Dropout and Deep Ensemble exhibit the highest QCDI despite their Bayesian motivation, that standard TS *inverts* quality-awareness (ρ flips from -0.033 to $+0.324$), and that QCTS reduces low-quality ECE by 68% while the architecture-level approach Q-VIB [2] remains the gold standard.

Code, benchmark data, and model weights will be released upon acceptance.

2 Related Work

Calibration and uncertainty estimation. Neural networks are notoriously miscalibrated [8]. Temperature Scaling learns a single post-hoc scalar and remains the dominant correction [8]. MC Dropout [7] and Deep Ensembles [12] estimate epistemic uncertainty but do not condition on input quality [14]. The Variational Information Bottleneck (VIB) [1] provides an information-theoretic alternative; Q-VIB [2] extends it by conditioning the prior on image quality scores.

Medical AI benchmarks and quality-aware deployment. Curated benchmarks [5, 9] prioritise discriminative accuracy on clean data; WILDS [11] examines in-the-wild distribution shift. IQA for dermoscopy has been studied with hand-crafted [4] and deep-learning approaches [17]; VisiScore-Net [3] serves as our quality backbone. Studies show performance degrades outside clinical settings [13, 21]. No existing work systematically evaluates how image quality affects calibration—the gap ITB fills.

3 Quality-Aware Calibration

3.1 Preliminaries

Let $f_\theta : \mathcal{X} \to \Delta^{K-1}$ be a classifier with predicted class $\hat{y} = \arg\max_k f_\theta(x)_k$ and confidence $\hat{p} = \max_k f_\theta(x)_k$. **ECE** [8] measures calibration via confidence bins:

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left| \mathrm{acc}(B_m) - \mathrm{conf}(B_m) \right|. \tag{1}$$

ECE is appropriate for i.i.d. test sets but fails when quality covariates create systematic variation. We additionally track predictive entropy $H(p) = -\sum_k p_k \log p_k$: for quality-aware models, entropy should increase as quality decreases.

3.2 QAC and QCDI

Let $q(x) \in [0,1]^5$ be the quality vector from VisiScore-Net [3] (sharpness, brightness, completeness, colour temperature, contrast) and $\bar{q}(x) = \frac{1}{5}\sum_i q_i(x)$ the scalar mean. We define **Quality-Aware Calibration (QAC)** as calibration conditional on quality stratum ℓ:

$$\mathrm{QAC}_\ell = \sum_{m=1}^{M} \frac{|B_m \cap \mathcal{Q}_\ell|}{|\mathcal{Q}_\ell|} \left| \mathrm{acc}(B_m \cap \mathcal{Q}_\ell) - \mathrm{conf}(B_m \cap \mathcal{Q}_\ell) \right|, \tag{2}$$

where $\mathcal{Q}_\ell = \{x : \bar{q}(x) \in [\tau_{\ell-1}, \tau_\ell)\}$. The **Quality-Calibration Degradation Index** summarises susceptibility in a single scalar:

$$\mathrm{QCDI} = \mathrm{QAC}_{\mathrm{low}} - \mathrm{QAC}_{\mathrm{high}}, \tag{3}$$

using ITB-LQ ($\bar{q} < 0.45$) and ITB-HQ ($\bar{q} > 0.50$) as strata. Positive QCDI indicates quality-oblivious calibration; negative QCDI indicates mild over-correction on low-quality images (the desirable outcome). Global ECE can be deceptively low even when QCDI is large: a model simultaneously underconfident on HQ and overconfident on LQ may show moderate aggregate ECE yet be dangerous in deployment.

3.3 Taxonomy of Calibration under Quality Shift

- **Quality-Oblivious** ($\mathrm{QCDI} \gg 0$). Calibration degrades sharply under quality shift. Uncertainty does not correlate with input quality. *Representative*: MC Dropout, Deep Ensembles (parameter-level uncertainty).
- **Quality-Fragile** ($\mathrm{QCDI} > 0$, moderate). Mild quality sensitivity—confidence partially decreases on degraded inputs—but insufficient to maintain calibration. *Representative*: Focal Loss + Label Smoothing (accidental quality sensitivity via training dynamics).
- **Quality-Aware** ($\mathrm{QCDI} \approx 0$ or negative). Calibration is stable across quality strata; confidence systematically decreases as quality degrades. *Representative*: Q-VIB (explicit prior conditioning), QCTS (post-hoc quality-dependent temperature).

This taxonomy is diagnostic: it reveals *why* a method fails and *what kind* of intervention is required.

4 Benchmark and Method

4.1 Image Triage Benchmark

ITB partitions a held-out ISIC 2020 test pool into four subsets by $\bar{q}$: **ITB-LQ** ($\bar{q} < 0.45$, $n = 300$, clinically critical degraded images); **ITB-Edge** ($\bar{q} \in [0.40, 0.55]$, $n = 660$, borderline quality); **ITB-HQ** ($\bar{q} > 0.50$, $n = 360$, clean images); and **ITB-Diverse** ($n = 1{,}500$, FitzPatrick17k across all six Fitzpatrick skin types).

Baselines. Ten methods: (A) EfficientNet-B3 (discriminative, no uncertainty); (D) Std VIB [1]; (E) Adaptive Prior VIB; (F) Q-VIB Full [2]; (G) Q-VIB+TokFT (discriminative variant); (H) Focal + Label Smoothing; (I) MC Dropout [7] (30 passes, p=0.2); (J) Deep Ensemble [12] (5 models); (TS) Temperature Scaling [8] applied to Std VIB; and (D + QCTS, ours).

Datasets. ISIC 2020 (33,126 images, 70/10/20 split) is the primary benchmark. HAM10000 [19 (10,015 images) and PAD-UFES [16] (2,298 images) serve as zero-shot transfer targets. Quality labels are obtained by applying VisiScore-Net to 149,100 programmatically degraded images across five dimensions.

4.2 Quality-Conditioned Temperature Scaling

Standard TS learns a uniform scalar T^* on $\mathcal{D}_{\text{val}}$:

$$T^* = \arg\min_T - \sum_{(x,y)\in\mathcal{D}_{\text{val}}} \log \text{softmax}(z(x)/T)_y, \tag{4}$$

applying the same temperature regardless of input quality. QCTS replaces T with a continuous function:

$$T(\bar{q}) = \text{softplus}(T_0 + \alpha \cdot (1-\bar{q})), \quad T_0, \alpha \in \mathbb{R}, \ \alpha \geq 0, \tag{5}$$

where $\text{softplus}(u) = \log(1+e^u)$ ensures $T > 0$. When $\bar{q}=1$, $T = \text{softplus}(T_0)$ (base temperature); as $\bar{q}$ decreases, T increases monotonically, softening predictions on degraded images. Setting $\alpha=0$ recovers standard TS.

(T_0, α) are optimised on $\mathcal{D}_{\text{val}}$ via L-BFGS (200 iterations) with the same NLL objective as standard TS:

$$(T_0^*, \alpha^*) = \arg\min_{T_0,\alpha} - \sum_{(x,y)\in\mathcal{D}_{\text{val}}} \log \text{softmax}\big(z(x)/T(\bar{q}(x))\big)_y. \tag{6}$$

QCTS is *post-hoc*: it requires no retraining, no architectural modification, and adds only two parameters. Unlike Q-VIB, which conditions its information bottleneck on $\bar{q}$ during training, QCTS corrects an already-trained model's confidence scale.

5 Experiments

Setup. All VIB models are trained on ISIC 2020 with AdamW (lr$=10^{-4}$, batch 128, 90 epochs, single RTX 4070). EfficientNet-B3 is fine-tuned with the same schedule. MC Dropout uses 30 stochastic passes at inference; Deep Ensemble comprises five independently initialised Std VIB models. TS and QCTS are optimised via L-BFGS on the validation split. We report AUC, ECE (M=15 bins), QCDI, and Spearman's $\rho(\text{entropy}, \bar{q})$ measured on HAM10000 zero-shot (to avoid stratification bias from ITB's design).

5.1 Main Results

Table 1 reports results for all ten methods. Three findings stand out.

Finding 1: High AUC does not imply quality-aware calibration. EfficientNet-B3 achieves the highest ITB-LQ AUC (0.751) yet the worst QCDI (0.276)—confirming that discriminative accuracy and quality-aware calibration are orthogonal objectives. MC Dropout

Table 1: **Main results on ITB.** Lower ECE and |QCDI| is better. Methods sorted by QCDI within taxonomy groups.

Method	LQ AUC	LQ ECE	HQ ECE	QCDI	ρ
Quality-Oblivious (QCDI > 0.10)					
A: EfficientNet-B3	0.751	0.345	0.070	+0.276	+0.251
I: MC Dropout	0.693	0.613	0.478	+0.135	-0.114†
J: Deep Ensemble	0.711	0.440	0.339	+0.101	-0.123†
Quality-Fragile ($0.04 < QCDI \leq 0.10$)					
G: Q-VIB+TokFT*	0.693	0.333	0.277	+0.056	-0.353
H: Focal + LS	0.708	0.535	0.492	+0.043	-0.401
Quality-Aware ($\lvert QCDI \rvert \leq 0.04$)					
D: Std VIB	0.553	0.146	0.128	+0.018	-0.033
D + TS	0.582	0.175	0.162	+0.013	+0.324
E: Adaptive Prior	0.588	0.149	0.138	+0.011	-0.151
F: Q-VIB Full	0.585	**0.149**	**0.140**	**+0.009**	**-0.164**
D + QCTS (ours)	0.563	**0.047**	0.062	**-0.015**	-0.108

ρ: Spearman corr. (entropy, $\bar{q}$) on HAM10000 zero-shot (more negative = quality-aware). † ITB-measured (subject to stratification bias). *Discriminative variant; supplementary. VIB models trade discriminative AUC for calibration via the information bottleneck.

(0.693 AUC, QCDI = 0.135) and Deep Ensemble (0.711 AUC, QCDI = 0.101) rank second and third worst. On ITB-LQ, MC Dropout's ECE reaches 0.613—four times that of Std VIB.

Finding 2: Standard calibration methods fail or actively worsen quality sensitivity. Temperature Scaling reduces ECE overall but *inverts* the entropy–quality relationship: ρ flips from -0.033 (Std VIB) to $+0.324$ (Std VIB + TS), meaning the TS-calibrated model exhibits *higher* entropy on high-quality images than on low-quality ones. This arises because TS uniformly softens logits, disproportionately affecting the confident predictions typical of high-quality inputs. Focal + Label Smoothing falls in the Quality-Fragile regime (QCDI = 0.043), not Quality-Oblivious as commonly assumed—implicit regularisation acquires some quality sensitivity, but insufficiently.

Finding 3: QCTS enters the Quality-Aware regime; Q-VIB is the gold standard. Q-VIB Full achieves QCDI = 0.009 and $\rho = -0.164$ (HAM10000), the strongest quality-aware behaviour overall. D + QCTS reduces ITB-LQ ECE by 68% ($0.146 \rightarrow 0.047$) and achieves QCDI = -0.015—the only post-hoc method to enter the Quality-Aware regime with maintained negative ρ. The gap ($\Delta\rho = 0.056$) quantifies the value of training-time conditioning. *Benchmark robustness:* QCDI rankings are stable across LQ thresholds $\tau_{LQ} \in [0.38, 0.46]$ (Kendall $\tau \geq 0.78$ for all adjacent pairs, mean 0.85), confirming that the taxonomy is not an artefact of threshold choice.

5.2 Calibration Taxonomy

Figure 2 visualises the taxonomy. EfficientNet-B3 (QCDI = 0.276) shows the starkest quality gap despite strong discriminative performance. Standard TS (QCDI = 0.013) appears Quality-Aware but its positive ρ (+0.324) reveals fundamentally oblivious behaviour—it softens all predictions uniformly, destroying quality–confidence correlation. Figure 3 shows diverging reliability curves for quality-oblivious methods vs. near-overlap for quality-aware ones.

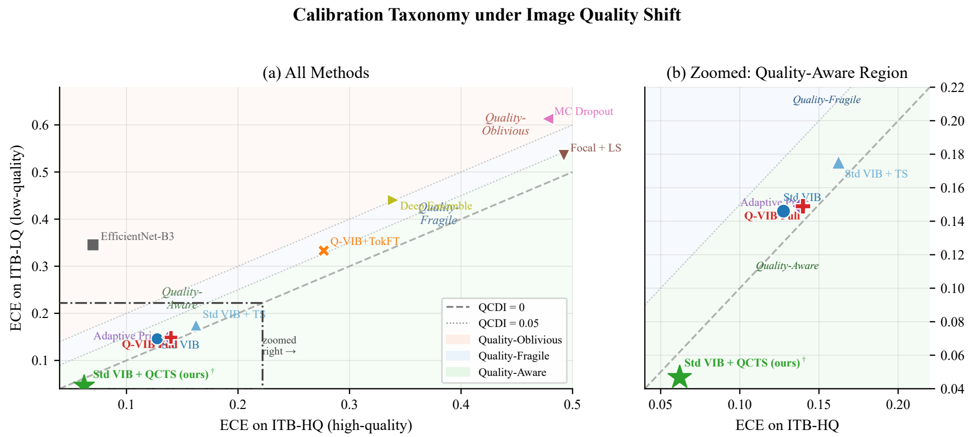

Figure 2: **Taxonomy scatter: ECE on ITB-HQ vs. ITB-LQ.** Distance above the diagonal equals QCDI; colour bands mark taxonomy class. D + QCTS and Q-VIB Full approach the $y{=}x$ (perfect invariance) line.

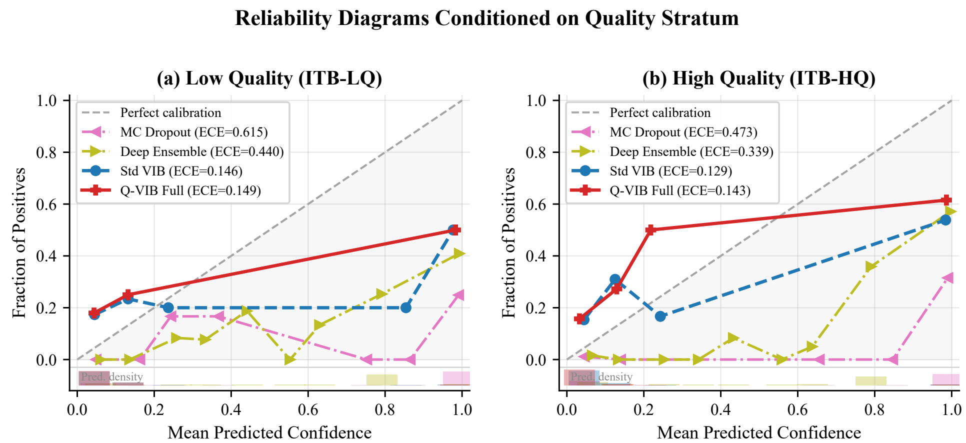

Figure 3: **Reliability diagrams conditioned on quality stratum.** Quality-oblivious methods (MC Dropout, Deep Ensemble) show diverging curves; quality-aware methods (Std VIB, Q-VIB Full) nearly overlap.

5.3 QCTS Analysis

Effect on calibration. Applying QCTS to Std VIB reduces ITB-LQ ECE from 0.146 to 0.047 (-68%), improves ρ from -0.033 to -0.108 (HAM10000), and transitions QCDI from $+0.018$ to -0.015 (slight over-correction on LQ). Figure 4 shows the learned $T(\bar{q})$ across three seeds.

Learned parameters. All three seeds converge to positive α (0.34, 0.96, 0.37), confirming that quality-dependent modulation is robustly discovered. The optimisation landscape has two apparent basins ($\alpha{\approx}0.35$ and $\alpha{\approx}0.96$); the best run ($\alpha{=}0.96$, $T_0{=}1.17$) yields $T(\bar{q}{=}0){=}2.24$ vs. $T(\bar{q}{=}1){=}1.44$—temperature nearly doubles for worst-quality images.

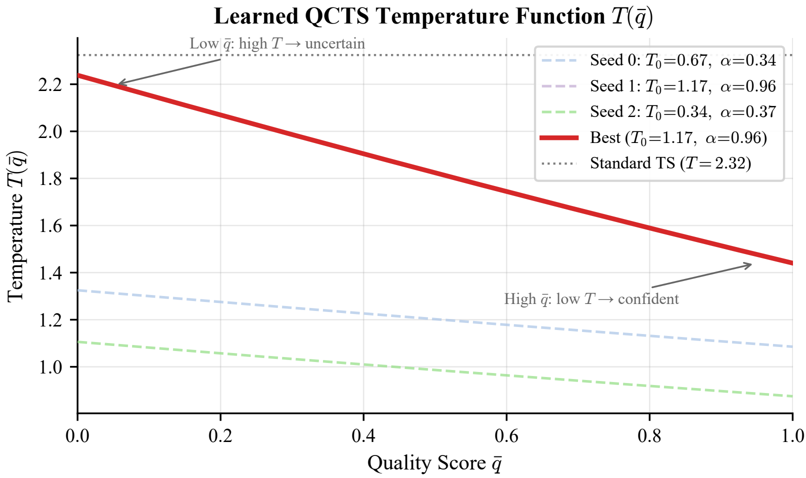

Figure 4: **Learned $T(\bar{q})$ across three seeds.** All converge to $\alpha{>}0$. Standard TS (horizontal dashed) is $\alpha{=}0$.

Boundary condition: QCTS requires quality-sensitive representations. Applying QCTS to EfficientNet-B3 yields $\alpha{\approx}0$, recovering standard TS. This null result is informative: QCTS amplifies quality-sensitive signals already present in a model's logits. Purely discriminative models produce logits uncorrelated with $\bar{q}$, leaving no signal to leverage. This establishes a key boundary condition for post-hoc quality-aware calibration.

Functional form ablation. We compare: (i) softplus QCTS (proposed, QCDI = -0.015); (ii) linear $T(\bar{q}){=}\max(T_0{+}\alpha(1{-}\bar{q}), 0.1)$ (QCDI = -0.004); and (iii) piecewise-constant T over $L{=}3$ quality bins (QCDI = $+0.029$). Despite piecewise achieving the lowest validation NLL (0.140 vs. softplus 0.152), it achieves the worst QCDI—demonstrating that *minimising NLL does not guarantee quality-aware calibration*. Smooth interpolation across the quality spectrum is essential.

5.4 Per-Degradation Analysis

Not all degradations damage calibration equally (Table 2, Figure 5). **Colour temperature** (q_4) is the most damaging across quality-oblivious methods (MC Dropout ECE = 0.627), followed by blur (q_1, ECE = 0.590). Colour temperature shifts confound model confidence in ways that misalign with accuracy, while blur—which destroys spatial detail— has a slightly

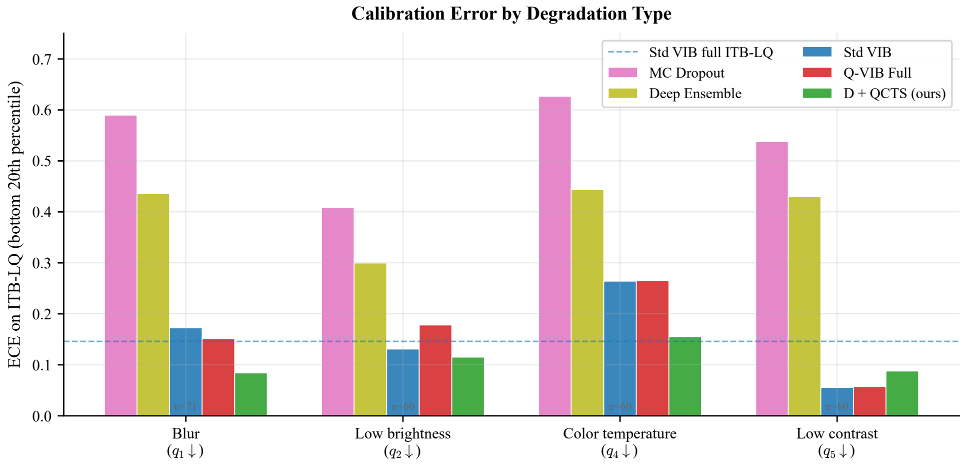

Figure 5: **ECE on ITB-LQ by degradation type** (bottom 20th percentile of each quality dimension). D + QCTS achieves the lowest ECE across three of four types.

weaker effect. Low contrast (q_5) has the weakest impact on Std VIB (ECE = 0.056) and Q-VIB (0.057)—VIB models are sensitive to feature magnitude rather than local contrast. D + QCTS achieves the lowest ECE across three of four degradation types; Std VIB is optimal for low contrast where QCTS over-corrects (0.088 vs. 0.056).

Table 2: **Per-degradation ECE on ITB-LQ** (bottom 20th percentile of each quality dimension). Bold marks best per row.

Degradation	MC Drop.	DeepEns.	Std VIB	Q-VIB	**D+QCTS**
Colour temp. ($q_4\downarrow$)	0.627	0.443	0.264	0.266	**0.155**
Blur ($q_1\downarrow$)	0.590	0.436	0.173	0.152	**0.084**
Low brightness ($q_2\downarrow$)	0.409	0.300	0.131	0.178	**0.115**
Low contrast ($q_5\downarrow$)	0.538	0.430	**0.056**	0.057	0.088

5.5 Cross-Dataset Generalization

To test generalisability, we evaluate all methods zero-shot on HAM10000 and PAD-UFES using VisiScore-Net quality scores (Figure 6). Q-VIB Full maintains significantly negative ρ on both: -0.164 ($p<10^{-60}$, HAM10000) and -0.236 ($p<10^{-29}$, PAD-UFES), consistent with ITB. QCDI rankings are significantly correlated between ITB and HAM10000 (Kendall τ=0.71, p=0.03). The correlation with PAD-UFES is weaker (τ=-0.43, p=0.24), because PAD-UFES images are clinical-quality with limited quality variation ($\bar{q}$>0.45 throughout), making absolute QCDI comparisons less meaningful. The entropy-based ρ metric, which does not require stratification, shows consistent patterns across all three datasets.

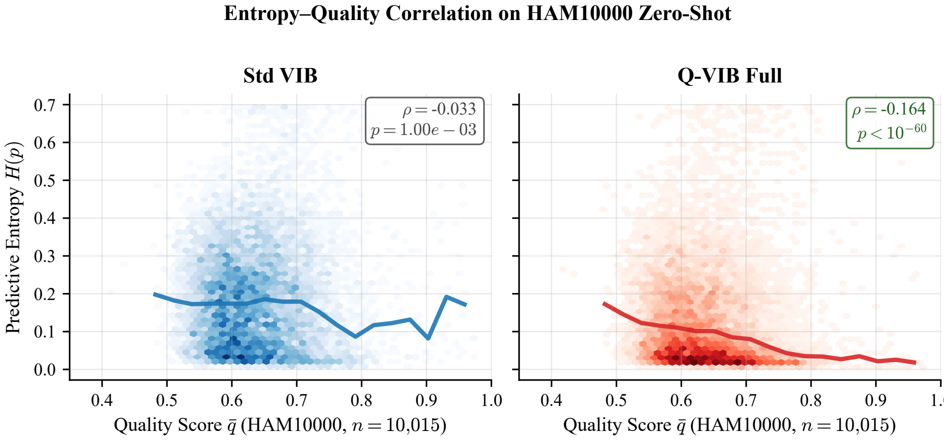

Figure 6: **Entropy vs. $\bar{q}$ on HAM10000 zero-shot** (n=10,015). Std VIB: near-zero correlation (ρ=−0.033). Q-VIB Full: strong negative trend (ρ=−0.164, $p<10^{-60}$).

6 Discussion

Why do standard methods fail? MC Dropout and Deep Ensembles inject uncertainty at the *parameter* level, independent of input quality: a blurry image and a sharp one produce identical dropout mask distributions. Temperature Scaling disproportionately softens large logits typical of HQ inputs, raising entropy *more* for high-quality images—inverting the quality–confidence relationship (ρ: $-0.033 \to +0.324$). Both failure modes share a common root: no mechanism conditions the predictive distribution on image quality. QCTS addresses this post-hoc by learning $T(\bar{q})$; Q-VIB addresses it architecturally. The QCTS–Q-VIB gap ($\Delta\rho$=0.056) quantifies the residual value of training-time conditioning.

Limitations and future work. ITB currently covers dermoscopy; extension to radiology or ophthalmology requires modality-specific quality models. Programmatic degradation may not capture all real-world artefacts (*e.g.* specular reflections or inconsistent gel). QCTS requires an external quality estimator, limiting applicability where none exists. Translating reduced QCDI to improved clinical decision support requires prospective evaluation.

7 Conclusion

We introduced Quality-Aware Calibration (QAC) and the Quality-Calibration Degradation Index (QCDI) to formalise and measure how model calibration degrades under image quality shift. Through the ITB benchmark—ten methods, four quality-stratified subsets, four datasets—we demonstrated that widely-used uncertainty methods are quality-oblivious despite their Bayesian motivation, that standard Temperature Scaling actively inverts quality-awareness, and that our proposed Quality-Conditioned Temperature Scaling (QCTS) reduces low-quality ECE by 68% post-hoc while the architecture-level Q-VIB remains the gold standard. A key boundary condition emerges: post-hoc quality-aware calibration requires the base model to carry quality-sensitive representations. We release ITB and QCTS to facilitate systematic quality-aware evaluation of medical AI.

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